Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Drawing Lines That Never Meet

Parallel lines are straight lines that stay the same distance apart all along their length, so they never meet, no matter how far you draw them. In a pair of parallel lines, the arrowheads drawn on each line point the same way.

To construct a line parallel to a given line through a chosen point, you can use two simple tools: a ruler and a set square. The set square is the triangle-shaped ruler; the secret is that its edges are already perfectly square to one another, so sliding it along a ruler keeps the same direction.

Here is the plan. You already have a line drawn on your page and a point P where you want the parallel line to pass. Put one edge of the set square along your given line. Hold your ruler firmly against another edge of the set square, like a track. Then slide the set square along the ruler until its edge touches the point P. Draw your line along that edge — it is parallel to the first line.

The important habit is to keep the ruler completely still while the set square slides. If the ruler moves, the direction changes and your new line will not be parallel.

Worked example — construct a line through a point P parallel to a given line AB.

Step 1: Draw a straight line and label its ends A and B. Mark a point P anywhere above the line where you want the new line to run.

Step 2: Place one edge of your set square flat along line AB. The set square should sit so it leans on the line.

Step 3: Hold your ruler against the long sloping edge of the set square, pressing it down so it cannot move. This ruler is your track.

Step 4: Slide the set square along the ruler until its bottom edge touches the point P. Do not let the ruler slip.

Step 5: Draw a straight line along the bottom edge of the set square, through P, and past it on both sides.

Step 6: Check your work — put the set square back on line AB and slide it again. If it lands exactly on your new line, you have drawn a true parallel. Mark both lines with matching arrowheads.

Practice

1. You are given a straight line. Use a ruler and a set square to construct a line parallel to it through a point P marked above the line. List the three things you must do, in order, to keep your new line parallel.
2. After constructing your parallel line, you slide your set square back along the ruler to check your work. What does it mean if the set square's edge lies exactly on the line you just drew?
3. While you are sliding the set square towards point P, your hand nudges the ruler and it turns slightly. Explain in one sentence why your new line will no longer be parallel.

Section 2: Drawing Lines That Meet at Right Angles

Two lines are perpendicular when they meet at a right angle — that is, the angle between them is exactly 90°. You can check your work with your protractor, but you can also construct a perpendicular line using only a ruler and a pair of compasses, which gives an exact right angle every time.

To construct a perpendicular, you only need two moves with your compass: draw arcs, then join the points where those arcs cross. Here is the whole idea. Suppose you have a straight line AB and you want to draw a line through a point P on it that stands straight up at a right angle. The secret is that the point where two equal arcs cross is always the same distance from the two centres of those arcs — and that balance is exactly what makes the join come out at 90°.

When you practise, always keep your compass opening unchanged as you swing each arc. If you keep the same opening, your arcs stay true and your right angle comes out cleanly.

Worked example 1 — a perpendicular at a point on the line

Construct a line perpendicular to AB through the point P that lies on AB.

Step 1. Put the point of your compass on P and open it to any width. Draw one arc that crosses AB on the left of P and another that crosses AB on the right of P. Call the two crossing points C (left) and D (right). Now PC = PD, because you used the same compass opening.

Step 2. Keep the compass at that same opening. Put the point on C and swing a large arc above P. Without changing the opening, put the point on D and swing another large arc above P. Let the two arcs meet at a point Q.

Step 3. Draw the straight line from P through Q. Because QC = QD and PC = PD, the line PQ stands exactly at a right angle to AB. Your protractor will read 90° at P.

Worked example 2 — a perpendicular dropped from a point not on the line

Construct a line from the point P that meets the line AB at a right angle, where P is not on AB.

Step 1. Put your compass on P and open it wide enough that an arc will cross AB in two places. Swing one arc that cuts AB at C on one side and D on the other.

Step 2. Now put the point of the compass on C and open it a little more than half of CD. Swing an arc below AB. Without changing the opening, put the point on D and swing another arc below AB. Let them meet at Q.

Step 3. Draw the line from P through Q. It crosses AB at a right angle, so the distance from P straight down to AB is the shortest possible. Check with your protractor: the angle where your new line meets AB is 90°.

Practice

1. Draw a horizontal line AB about 6 cm long and mark a point P in the middle of it. Using only a ruler and compasses, construct the line through P that is perpendicular to AB. What is the angle where your new line meets AB?
2. A point P is drawn above a line AB, not touching it. You want to drop a perpendicular from P down to AB. Put the steps in order: (a) swing an arc centred at P that cuts AB in two places; (b) draw the line from P through the meeting point of the two lower arcs; (c) from each cutting point, swing equal arcs that meet below AB. Write the correct order as three letters.

Section 3: Cutting a Line Segment in Half with a Compass

When you "bisect a line segment", you cut it exactly into two equal halves. The point where it is cut is called the midpoint. You can find that midpoint with a ruler, but a much more reliable way is to do it by construction with a compass — and that is what this revision pack is about.

The idea is simple. If you draw a compass circle that passes through both ends of your segment, then the point halfway between those two ends is exactly the centre of that circle. To be sure, we draw two overlapping circles (one centred at each end), and the midpoint of the segment lies on the straight line joining the two points where the circles cross.

Here is the full method, step by step. You need only a ruler (for drawing straight lines) and a pair of compasses.

Step 1. Draw your line segment AB.
Step 2. Put the point of the compass on A, open it a little more than half of AB, and draw a large arc (or circle) so it swings well above and well below the line.

Step 3. Keeping the compass opening exactly the same, put the point on B and draw another arc, the same size. The two arcs should cross each other at two points — one above the line and one below it.
Step 4. Draw a straight line through those two crossing points. Where this new line cuts AB is the midpoint of AB.

The new line you have drawn is called the perpendicular bisector of AB, because it crosses AB at a right angle and at its exact middle. The single point where the two lines meet is the midpoint.

Worked example: Bisect a segment 8 cm long.

Suppose AB is 8 cm long and you want its midpoint.

1. Draw AB, 8 cm long, with A on the left and B on the right.
2. Open your compass to about 5 cm — clearly more than half of 8 cm (half of 8 cm is 4 cm, so 5 cm works). Put the point on A and swing a large arc above and below the line.
3. Keep the compass opening at 5 cm. Put the point on B and swing a matching arc. The two arcs meet at a point above the line, call it C, and a point below the line, call it D.
4. Draw a straight line through C and D. It will cross AB at the midpoint M.
5. Measure: AM should be 4 cm and MB should be 4 cm. Because the opening (5 cm) was bigger than half the segment, this works even if you do not measure AB first.

Check: the two halves AM and MB are equal, and the line CD meets AB at a right angle. You have bisected the segment.

Practice

1. A line segment PQ is 10 cm long. Using your compass and ruler only, bisect PQ and mark the midpoint M. What is the length of PM?
2. A segment XY is 6 cm long. You bisect XY to find its midpoint, then you bisect one of the two halves again. How long is each of the smallest pieces?

Section 4: Drawing a Perfect 90° Angle

A right angle measures exactly 90°. It is the corner you see in a square or the corner of a sheet of paper. Using only a ruler, a pencil and a pair of compasses, you can draw a perfect 90° angle without guessing.

Here is the method to construct a 90° angle at a chosen point P on a straight line. Keep your pencil sharp, and do not change the compass opening unless the steps tell you to.

1. Draw a straight line and mark a point P on it.
2. With P as the centre, draw an arc (a part of a circle) that cuts the line at two points, one on each side of P. Call them A and B.
3. Open the compass a little wider — more than half the distance from A to B.
4. With A as the centre, draw an arc above the line.
5. Keeping the compass exactly the same opening, with B as the centre, draw another arc above the line to cut the first arc. Call the point where they meet C.
6. Use your ruler to join P to C, and extend the line.

The line PC meets the original line at right angles. The two lines are perpendicular, so the angle between them is exactly 90°.

Why does this work? The arcs drawn from A and B have equal radius, so C is the same distance from A as it is from B. That means C sits exactly above the middle of the segment, and the line joining it to P stands straight up — at right angles to the line below.

Construct a 90° angle at the point P on the line below.

Step 1: Draw a straight line and mark a point P somewhere on it.

Step 2: Put the point of your compass on P and draw an arc that crosses the line on both sides of P. Label these crossings A (left) and B (right). Notice that A and B are the same distance from P.

Step 3: Widen the compass so its opening is more than half of AB. Put the point on A and draw an arc above the line.

Step 4: Without changing the compass opening, put the point on B and draw another arc above the line. The two arcs meet at a point C.

Step 5: Place your ruler against P and C, and draw a straight line through both points, extending it a little past C.

The line PC now stands at right angles to the original line, so the angle at P is exactly 90°. To check, you can fit the corner of your set square or a sheet of paper into the angle — it should sit perfectly.

Practice

1. List the compass and pencil steps, in order, that you would use to construct a 90° angle at a point P on a straight line.
2. In the construction of a 90° angle, you draw two arcs above the line — one from centre A and one from centre B. Why must the compass opening be the same for both of these arcs?
3. A student marks a point P on a line and draws the first arc with centre P, getting points A and B. She then keeps the same small compass opening to draw arcs from A and B, but the two arcs do not meet above the line. What has she done wrong, and how should she fix it?
4. Using a ruler, a pencil and a pair of compasses, construct a 90° angle at a point P on a straight line. Then use the corner of your set square or a sheet of paper to check that the angle you drew is exactly 90°.

Section 5: Constructing a 60° Angle

An angle of 60° is one of the special angles you can draw using only a ruler and a pair of compasses. You do not need a protractor at all. The trick is that a 60° angle is one corner of an equilateral triangle — a triangle in which all three sides are equal, so all three angles are equal too. Since the three angles of any triangle add up to 180°, each angle in an equilateral triangle is $180° \div 3 = 60°$.

So, to draw a 60° angle, you simply build an equilateral triangle. Here is the plan:

1. Draw a straight line, and mark a point on it. Call this point A. This is where your angle will sit.
2. Open your compasses to any convenient length. Keep this opening the SAME for the whole construction.
3. Put the compass point on A and draw an arc that crosses the line on one side of A. Call the crossing point B. The distance AB is your compass opening.
4. Keep the compasses at the same opening. Put the point on B and draw an arc that crosses the first arc (above the line). Call the crossing point D.
5. Draw a straight line from A through D.

The angle between the line AB (the original line) and the line AD is exactly 60°. You have constructed a 60° angle! You can check why it works: D lies on the arc drawn from A, so AD equals your compass opening; D also lies on the arc drawn from B, so BD equals your compass opening. And AB equals your compass opening too. So triangle ABD has all three sides equal — it is equilateral — and its angle at A is 60°.

Worked example: Construct a 60° angle at point A on a line.

Step 1 — Draw a horizontal line and mark point A on it.

Step 2 — Open your compasses to a convenient width, say 4 cm. Put the compass point on A and draw an arc that cuts the line on the right. Where it cuts the line, mark point B. So AB is now 4 cm long.

Step 3 — Keep the compasses at exactly 4 cm. Put the point on B and draw an arc above the line so that it crosses the first arc. Mark the crossing point D.

Step 4 — Draw a straight line from A through D. The angle DAB is 60°.

Check: AB is 4 cm (your compass opening). Point D lies on the arc drawn from A, so AD is also 4 cm. Point D also lies on the arc drawn from B, so BD is 4 cm as well. All three sides of triangle ABD are equal, so it is equilateral. That means each of its angles is 60°, so the angle at A is 60°.

Practice

1. Construct a 60° angle at a point A on a straight line using only a ruler and compasses. Then measure it with your protractor (if you have one) to check it is about 60°.

2. In your construction of a 60° angle, the point where the two arcs cross above the line is labelled D. Why is the angle at A exactly 60° and not some other size? Explain using the idea of a triangle.

Check your work

Attempt every question before you read this page.

Drawing Lines That Never Meet

1. a. Place one edge of the set square along the given line.
   b. Hold the ruler firmly as a track against another edge of the set square so it cannot slip.
   c. Slide the set square along the ruler to P and draw along its edge.

   The three essential steps are: put the set square's edge on the given line; lock the ruler against the set square as a track; slide the set square to P and draw. The ruler must stay still while the set square slides, otherwise the direction changes.
2. It means the line you drew is correctly parallel to the given line, because sliding the set square along the same track reproduced the same direction.

   Sliding the set square along the ruler reproduces the same direction every time, so if it lies exactly on your new line, that line has the same direction as the original — it is parallel. If it does not land exactly, the ruler moved during construction and you should try again.
3. Because the ruler is the track that fixes the direction, so if it turns, the set square slides in a new direction and the line you draw is no longer parallel to the original.

   The ruler sets the direction of the slide. The whole method only works because sliding the set square along a still ruler preserves the original direction. If the ruler turns, the preserved direction changes too, so the drawn line is not parallel.

Drawing Lines That Meet at Right Angles

1. The angle is exactly 90° (a right angle), so the constructed line is perpendicular to AB.

   Put the compass point on P and draw two small arcs, one crossing AB on each side of P. Keep the same opening, put the point on each crossing in turn, and swing two larger arcs that meet above P. Draw the line from P through that meeting point. The new line makes a 90° angle with AB, so it is perpendicular.
2. a, c, b

   First the arc from P cuts AB in two places (a). Then, from those two places, equal arcs meet below the line (c). Finally you join P to the meeting point (b). The order is a, c, b.

Cutting a Line Segment in Half with a Compass

1. PM = 5 cm, because the midpoint splits PQ into two equal halves of 10 ÷ 2 = 5 cm each.

   Draw PQ = 10 cm. Open the compass to more than half of 10 cm (for example, 6 cm). Draw an arc from P above and below the line, then, keeping the same opening, draw an arc from Q. The arcs cross at two points; draw a straight line through both crossing points. Where that line meets PQ is M. Since M is halfway along a 10 cm segment, PM = 10 ÷ 2 = 5 cm.
2. The smallest pieces are 1.5 cm each.

   First bisection splits XY = 6 cm into two equal pieces of 6 ÷ 2 = 3 cm each. Bisecting one 3 cm piece splits it into two equal pieces of 3 ÷ 2 = 1.5 cm each. So the smallest pieces are 1.5 cm long.

Drawing a Perfect 90° Angle

1. a. Put the compass point on P and draw an arc cutting the line at two points A and B.
   b. Widen the compass to more than half of AB.
   c. Draw an arc above the line from centre A.
   d. Keeping the same opening, draw an arc above the line from centre B to meet the first arc at C.

e. Draw the line from P through C. The angle at P is 90°.

The construction needs three centres: first P to mark A and B, then A and B to locate C. After the first arc fixes A and B, open the compass wider so the arcs from A and B cross above the line. The meeting point C is then joined to P.

2. So that the point C where the arcs meet is the same distance from A as it is from B. That places C directly above the middle of the line, making the angle at P exactly 90°.

   Equal radii from A and B mean C lies on the perpendicular bisector of AB, so the line PC is perpendicular to the line, giving a 90° angle. If the openings differed, C would not be centred and the angle would not be exactly 90°.

3. Her compass opening is too small — the arcs do not reach each other. She should widen the compass so its opening is more than half of AB, then draw the arcs from A and B again.

   The first arc from P may be any size, but the arcs from A and B must be opened wider than half of AB, otherwise the two arcs have too small a radius to cross above the line. Widening the compass fixes it.

4. The construction should show a straight line, a point P on it, and a second line through P standing at right angles to the first. The corner of a set square or paper should fit into the angle exactly.

   Follow the five steps: mark P, arc from P to get A and B, widen the compass, arc from A, arc from B to meet at C, then join P to C. Check the corner of a set square fits snugly in the angle, confirming 90°.

Constructing a 60° Angle

1. You should end with an angle between your original line and the final line of about 60°. The two arcs above the line cross at a point D, and the line from A to D makes the 60° angle.

   Draw a line and mark A. With compasses at a fixed opening, mark B on the line from A, then C on the line from B. Keeping the same opening, draw arcs from B and C that cross above the line at D. Draw A to D. Since AB = BC = AD (all equal to your compass opening), triangle ABD is equilateral, so the angle at A is 60°.

2. Because triangle ABD is equilateral: AB, BD and AD are all the same length (each equals your compass opening). Every angle of an equilateral triangle is 180° ÷ 3 = 60°, so the angle at A is 60°.

   All three sides of triangle ABD are the same length, so it is equilateral. The three equal angles add to 180°, so each is 60°. The angle at A is one of them.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-8c16ff29	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____